



The Expression of Ego-Depletion

Thin Slices of Nonverbal Behavior as Cues to Momentary Self-Control Capacity

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Abstract: Is depleted self-control capacity detectable from thin slices of behavior? In four video-conditions (early-depleted; early-non-depleted; late-depleted; late-non-depleted) untrained observers rated target persons' available self-control capacity in Experiments 1 and 2 (without sound) and in terms of fatigue and negative affect (Experiment 3). Videos were analyzed with Noldus FaceReader software (Experiment 4). Ego-depletion could reliably be detected from thin slices of behavior. Nonverbal expressions coinciding with ego-depletion are associated with cues signaling momentary levels of self-control capacity, fatigue, and some negative affective states. FaceReader analyses indicated that facial expressions coinciding with ego-depletion are subtle and are extractable from facial dynamics rather than expression intensities. Results indicate that self-control depletion might not only have intrapersonal effects, but also interpersonal effects.

Keywords: person perception, social cognition, self control, nonverbal behavior/communication, facial expressions

Walter Mischel's legendary "Marshmallow Test" (Mischel, Shoda, & Rodriguez, 1989) highlighted the importance of self-control on success in life (for a critical discussion, see Watts, Duncan, & Quan, 2018). People familiar with the "Marshmallow Test" probably have a vivid image of a young child in their head trying to control itself to gain a larger reward later (e.g., two marshmallows) while confronted with the immediate temptation of having one, highly desired, marshmallow right away. When watching the little children struggle to control themselves one can see their internal struggles intensify in their observable behavior over time (Mischel, 2014). Beyond anecdotal evidence, however, there are hardly any empirical data to support the common experience that exerting self-control may coincide with specific nonverbal behavior in the face and the body.

Self-control has been defined as the process of modifying one's own predominant behavioral, emotional, and cognitive responses according to a given standard (e.g., goals, norms, values, morals, laws; Baumeister & Vohs, 2016). When people's self-control capacities are relatively low due to their dispositions or situational circumstances, they are less likely to suppress their impulses and behave according to social norms. For instance, in their influential work, Gottfredson and Hirschi (1990) concluded that a lack

of self-control is an important determinant of delinquent and antisocial behavior (see Pratt & Cullen, 2000, for a meta-analysis on the corresponding empirical support). In addition, it has been found that momentary reduced self-control capacity in undergraduates made them more likely to aggress toward provocateurs (i.e., harm them intentionally; DeWall, Baumeister, Stillman, & Gailliot, 2007; Stucke & Baumeister, 2006). On the other hand, people with a relatively high self-control capacity tend to behave prosocial. For example, they are more likely to assist others who need help (DeWall, Baumeister, Gailliot, & Maner, 2008). Hence, self-control plays a crucial role in social functioning in communities.

Although most people would agree about the importance of self-control in social life, a lively debate has emerged due to controversies on theoretical conceptions (Lurquin & Miyake, 2017) and empirical findings (Hagger et al., 2016) pertaining to self-control. The debate mainly focuses on the strength model of self-control (Baumeister, Vohs, & Tice, 2007) that compares self-control to the functioning of a muscle. Multiple experimental studies (Hagger, Wood, Stiff, & Chatzisarantis, 2010; see Inzlicht, Gervais, & Berkman, 2015 for a current discussion of this literature) have argued that the initial use of the "self-control muscle"

will cause a decrease in self-control strength when needed again on a subsequent task – a state that has been termed *ego-depletion* (Baumeister, Bratslavsky, Muraven, & Tice, 1998). To investigate the ego-depletion effect, the so-called sequential-task paradigm is typically applied: Participants are assigned to a group that works either on a primary task requiring self-control (i.e., ego-depletion condition) or on a task that does not require self-control (i.e., control condition). This should lead to a state of ego-depletion in the former condition, while participants' self-control strength from the latter condition should remain relatively intact. However, increasing research findings have cast doubt on the existence of a limited resource fueling self-control (e.g., Inzlicht & Schmeichel, 2012; Job, Dweck, & Walton, 2010; Kurzban, Duckworth, Kable, & Myers, 2013), resulting in alternative explanations for the ego-depletion effect, including affective, motivational, and cognitive processes. Guided by evolutionary considerations, the ego-depletion effect has alternatively been described as a product of evolutionary pressures that motivate the individual to balance effortful (but necessary) obligations and desirable exploration (Inzlicht, Schmeichel, & Macrae, 2014). According to this model, self-control failure is not about resource depletion, but about switching priorities from "have-to" to "want-to" goals. In an updated version of the strength model, Baumeister and Vohs (2016) still stick to the idea of a limited self-control resource; however, they incorporate several suggestions from the newly emerging literature and now suggest that a central governor selectively allocates resources in response to executive demands. While this updated model can account for more of the newly emerging findings in the ego-depletion literature, it has been criticized for being a too flexible model that cannot be falsified (Lurquin & Miyake, 2017).

For the present research, it is not of central importance which process (e.g., limited strength resource or motivational shift) underlies the ego-depletion effect. We rather examine whether lowered self-control capacity as the result of such a mechanism – indicated by performance decrements after an initial self-control task – is communicated nonverbally. Regarding this research question, the motivational shift model by Inzlicht and colleagues is arguably more useful as it predicts that it is not so much that people cannot control themselves in the sequential-task paradigm, but instead do not feel like controlling themselves and prefer to indulge in more inherently enjoyable and easier task pursuits. Thus, depleted individuals are assumed to experience changes in motivation and affect (Inzlicht et al., 2014) and, importantly, this should be more pronounced the more self-control effort people have already invested (Lindner, Nagy, Ramos Arhuis, & Retelsdorf, 2017).

Of further relevance to the present research, a large body of evidence suggests that motivational and affective

internal states are detectable from a persons' nonverbal behavior (see Ambady, Bernieri, & Richeson, 2000 for a review). Several theoretical accounts predict that humans are highly efficient at communicating internal states (Bogdan, 1997) like emotions (Darwin, 1872; Ekman, 1992) or social intentions (Fridlund, 1994) nonverbally. Although the emotions view (Basic Emotion Theory; Ekman, 1992) and the social intentions view (Behavioral Ecology Theory; Fridlund, 1994) are rival theories on several accounts and have led to vivid debate (Crivelli & Fridlund, 2018; Ekman, 2016b), they both share the tenet that humans have evolved to be highly sensitive to the facial displays of others and are (often) able to draw accurate inferences from these displays. Pertinent to the present research, the Behavioral Ecology Theory suggests (in contrast to Basic Emotion Theory) that nonverbal (e.g., facial) displays do not necessarily have to be functional to the individual showing the display. That is, nonverbal displays may be intentional or unintentional and are believed to guide further social interaction with observers – for the better or the worse (Crivelli & Fridlund, 2018; Fridlund, 1994). In support of this theorizing, previous research has for example shown that trailing (i.e., currently losing) athletes display nonverbal behavior that observers accurately interpret as signs of losing (Furley & Schweizer, 2014a), although this is arguably not functional for the trailing athletes (Furley & Schweizer, 2014b). Similarly, soccer referees have been shown to "leak" nonverbal signs of insecurity when communicating relatively difficult decisions in comparison to relatively easy decisions (Furley & Schweizer, 2016a).

To determine the extent to which observers are able to draw meaningful inferences from the nonverbal behavior of others, researchers often use the *thin-slices paradigm* (see Ambady et al., 2000; Ambady & Rosenthal, 1992 for an overview). Within the *thin-slices paradigm*, observers watch short video-recordings or images of target persons and are asked to form judgements about the target persons' internal states or to predict a likely outcome of a social interaction. These observer judgments or estimates are subsequently compared with reliable measurements of the variables in question. Empirical research has demonstrated that people are often highly accurate in their judgments: for example, about mental states (Baron-Cohen et al., 1996), conversation outcomes (Curhan & Pentland, 2007), teaching effectiveness (Ambady & Gray, 2002), leadership competence (Rule & Ambady, 2008), and sales effectiveness (Ambady, Krabbenhoft, & Hogan, 2006).

People's general ability to detect others' internal state on the basis of nonverbal cues suggests that if the act of exerting self-control is associated with internal processes and changes, observers should be able to recognize cues associated with these internal processes and changes. From the perspective of the perceiver, the ability to notice vis-à-vis

other's general readiness to be impulsive (low self-control capacity) or considerate (high self-control capacity) can serve adaptive functions at the level of individual goal attainment, for example, who should be avoided in a confrontational situation or who should be approached for help (Fridlund, 1994; McArthur & Baron, 1983; Zebrowitz & Collins, 1997). Thus, it seems reasonable to assume that people are able to notice others' state of self-control capacity via nonverbal behavior (e.g., in the face or body), as this would allow to roughly anticipate likely behavior toward oneself and to adapt oneself accordingly (Fridlund, 1994).

Following from the above, the present research addresses the question if ego-depletion (i.e., a lowered readiness for self-control) is detectable from thin slices of behavior (Ambady et al., 2000; Ambady & Rosenthal 1992). Specifically, we predicted that even brief exposures to individuals' faces permit an accurate estimation of their state self-control capacity. As self-control capacity declines progressively (Lindner, Nagy, Ramos Arhuis, & Retelsdorf, 2017), we further assumed that reduced self-control capacity should be more pronounced in a person's nonverbal behavior after a prolonged period of self-control exertion than at an early stage of self-control exertion. We tested this hypothesis in four studies by initially filming participants (preliminary study) in a common version of the sequential task paradigm (Baumeister et al., 1998) and selecting short video recordings from this experimental protocol that observers were asked to rate in terms of momentary self-control capacity (Experiments 1 and 2), fatigue, and negative affect (Experiment 3). Finally, Experiment 4 sought to explore which objective cues people might use to determine a person's level of self-control capacity by analyzing the video sequences with Noldus FaceReader.

Experiment 1

Preliminary Study

In an initial step of creating the stimulus material, we used a common version of the sequential task paradigm (Baumeister et al., 1998) while filming the participants' faces and upper bodies with a web-cam (C920 Logitech HD Pro, Logitech, Lausanne, Switzerland) during the entire time of the computer-based protocol. In line with recommendations from the thin slices literature (Carney, Colvin, & Hall, 2007; Murphy et al., 2014) we attempted to recruit 40 different participants from which to sample

video sequences showing thin slices of behavior during the experimental protocol.¹ Due to sampling and technical issues, we only had 36 participants for which all necessary video data were available (full details to this preliminary study can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d). Participants gave their written informed consent for using this video material for the purposes of this research study.

First, participants were asked to watch an emotional video for ten minutes containing both unpleasant scenes (e.g., sports injuries) and pleasant scenes (e.g., funny clips). One group (depletion group) was asked to suppress their emotional reactions while watching, whereas the other group was asked to watch the video as they typically would (non-depletion group; for this procedure, see Muraven, Tice, & Baumeister, 1998). Recently, based on meta-analytic findings, Dang (2018) concluded that this kind of ego-depletion induction is the most effective one (see also Wolff, Baumann, & Englert, 2018). Afterward, participants responded to a series of typical control and manipulation check measures verifying that the task required relatively more self-control in the depletion condition than in the non-depletion condition.

Subsequently, participants performed the Color-Stroop task (Stroop, 1935), in which participants indicated the font color of a series of color words appearing subsequently on the computer screen. On congruent trials, the font color and the semantic meaning of the color word matched ("red" written in red font color), while on incongruent trials there was a mismatch between the two (e.g., "red" written in blue font color). While incongruent trials require self-control, as individuals have to volitionally change their dominant tendency to name the font color instead of reading the actual color word, congruent trials do not require self-control. Previous research has shown, that depleted individuals perform significantly worse on incongruent trials as they have relatively less self-control capacity to support task performance (e.g., Wallace & Baumeister, 2002).

Results of our initial study showed that participants in the depletion group indeed exerted more self-control in watching the emotional videos (see Electronic Supplementary Material, ESM 1). Importantly, depleted participants further performed significantly worse on the Stroop test than non-depleted participants. Specifically, the ego-depleted group showed more interference (standard Inquisit interference measure = incongruent trial [e.g., the word green printed in blue] minus the neutral trials [e.g., the word dog printed in blue]) on the Stroop Task compared to the non-depleted

¹ The preliminary study was conducted during a weekend university seminar and therefore the video clips had to be filmed in the facilities available. Although this added "noise" to the stimulus material, we made sure that the experimental categories were not confounded. We did not attempt to recruit more participants as the study protocol of the main studies of rating thin slices of participants' recorded nonverbal behavior would have become too long.

group, $t(30, 1) = -2.022$, $p = .025$, one-tailed, $d = 0.68$; non-depletion $M = 177.65$, $SD = 73.44$ versus ego-depletion $M = 237.92$, $SD = 102.92$. These results suggest that ego-depleted participants had more difficulty suppressing an inappropriate, preponent response tendency due to their momentary lack of self-control capacity (see ESM 1 for more details on this initial study; and https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d).

Participants

To examine observers' perceptions of the preliminary study's participants, we attempted to recruit at least 55 participants for the main study, which would provide sufficient power (.95) to obtain small-to-medium effects ($f = 0.2$) in the 2 (Measurement Time) $\times$ 2 (Depletion Condition) within-subject design (Faul, Erdfelder, Lang, & Buchner, 2007). Participants were recruited and tested online. Data collection was terminated after reaching 57 complete datasets ($N = 57$, 34 male and 23 female; $M_{\text{years}} = 24.89$; $SD = 4.87$). All participants gave written informed consent.

Procedure

Based on the initial study and recommendations from the thin slices literature (Carney et al., 2007; Murphy et al., 2014), we created 10-second slices of participants performing the Stroop task (i.e., naming the font color of a color word instead of its semantic meaning, e.g., green) at two different times: the first 10 s after the Stroop practice trials (early condition) and the last 10 s of the Stroop task (late condition). The rationale for this experimental factor was based on research showing that self-control capacity declines progressively (Lindner et al., 2017) and therefore reduced self-control capacity should theoretically show more in a person's nonverbal behavior after a prolonged period of self-control exertion than at an early stage of self-control exertion. As a result, we had a total of 72 videos in 4 experimental conditions (18 clips respectively): early-depleted, early-non-depleted, late-depleted, and late-non-depleted. The experimental protocol was administered online via SoSci Survey (Leiner, 2014). Importantly, the videos contained visual nonverbal information (i.e., participants' facial and bodily expression) as well as auditory information (i.e., participants' pronouncing the color words). All clips were shown in different random orders and participants were asked to respond to seven items on scales ranging from 1 to 8 (higher ratings always represented higher self-control capacity) derived from an established measure to assess momentarily available

self-control capacity (SSCCS-K-D; Bertrams, Unger, & Dickhäuser, 2011): How does the target person appear to you?

- (1) awake [wach] and concentrated [konzentriert];
- (2) balanced [ausgeglichen];
- (3) lack of drive [antriebslos];
- (4) exhausted [erledigt];
- (5) mentally fatigued [geistig erschöpft];
- (6) depleted [ausgelaugt];
- (7) in terms of willpower [Willenskraft].

These items were based on previous *thin-slices* research (Furley & Schweizer, 2016b; Rule, Adams, Ambady, & Freeman, 2012) on nonverbal behavior and purposefully modified for this experiment. All study materials can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

Results

The descriptive statistics are summarized in Table 1.

A 2 (Early vs. Late) $\times$ 2 (Depleted vs. Non depleted) ANOVA on the mean ($M_{\text{early-depleted}} = 5.44$, $SD = 0.65$; $M_{\text{early-non-depleted}} = 5.58$, $SD = 0.68$; $M_{\text{late-depleted}} = 4.73$, $SD = 0.74$; $M_{\text{late-non-depleted}} = 5.04$, $SD = 0.72$) of the seven dependent variables ($\alpha = .97$) revealed a significant main effect of measurement time (early vs. late) on overall perceived self-control of the targets, $F(1, 56) = 171.744$, $p < .001$, $\eta_p^2 = .754$. Participants rated the targets substantially lower on perceived self-control toward the end of the Stroop task as compared to the beginning of the Stroop task. A significant main effect of depletion group, $F(1, 56) = 32.689$, $p < .001$, $\eta_p^2 = .369$, indicated that participants rated targets as possessing less perceived self-control strength when targets belonged to the depletion group as compared to when targets belonged to the non-depletion group. The interaction between measurement time and depletion group was also significant, $F(1, 56) = 5.489$, $p < .05$, $\eta_p^2 = .089$, demonstrating that the effect of time was more pronounced in the depletion group as the difference in perceived self-control strength was larger between the two experimental groups toward the end of the Stroop task as compared to the beginning of the Stroop Task (see Table 1). Further analyses on the individual perceived self-control items can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

Discussion

Experiment 1 delivers initial empirical evidence that observers rated targets lower on items that have been linked to

Table 1. Descriptive statistics (*M* and *SD* in brackets) as a function of measurement time and depletion group on the individual dependent variables in Experiment 1 (with sound). Low values always reflect low levels of self-control capacity on the individual items

	Early		Late	
	Depleted	Non-depleted	Depleted	Non-depleted
Awake (1 = <i>not at all</i> ; 8 = <i>very</i>)	5.70 (0.67)	5.81 (0.73)	4.94 (0.86)	5.33 (0.82)
Balanced (1 = <i>not at all</i> ; 8 = <i>very</i>)	5.43 (0.76)	5.46 (0.77)	4.94 (0.86)	5.07 (0.82)
Lack of drive (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.35 (0.72)	5.65 (0.68)	4.55 (0.80)	5.03 (0.78)
Exhausted (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.33 (0.70)	5.45 (0.76)	4.49 (0.84)	4.80 (0.81)
Mentally fatigued (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.44 (0.77)	5.50 (0.80)	4.69 (0.86)	4.91 (0.90)
Depleted (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.35 (0.72)	5.48 (0.80)	4.55 (0.89)	4.83 (0.86)
Will power (1 = <i>not at all</i> ; 8 = <i>very</i>)	5.46 (0.66)	5.72 (0.65)	4.97 (0.70)	5.29 (0.64)

temporally available self-control capacity (Bertrams et al., 2011) after prior exertion of self-control. The observers were naïve about what the target people did before each of the 10-second clips: that is, participants neither explicitly knew whether the target persons executed self-control or not while watching the initial emotional video, nor did they know at what point the target person was in the Stroop task (at the beginning or the end of the Stroop task). The results further supported the hypothesis that reduced self-control capacity showed more in a person's nonverbal behavior after a prolonged period of self-control exertion (late condition) than at an early stage of self-control exertion (early condition).

The targets nonverbal behavior in the face, upper body, and their voice (i.e., naming the Stroop colors) was the only information available to estimate the targets' state of self-control capacity. As predicted, the participants' ratings corresponded to the targets' state of ego-depletion. This was even true for the initially non-depleted control group of targets, indicated by the drop in perceived self-control capacity during the second self-control task (i.e., the Stroop task) – though, as expected, this effect was stronger for the initially depleted target group.

Experiment 1 left unexplained, however, which nonverbal cues people use to assess others' self-control capacity. In addition to bodily and facial expressions, the participants also heard the tone of voice and potential cues such as hesitation while the target people named the Stroop colors. Therefore, in Experiment 2, we examined whether bodily and facial expressions are sufficient to perceive different levels of self-control capacity in others.

Experiment 2

Experiment 2 was identical to Experiment 1 only that the videos were presented without sound. The purpose of this was to investigate if ratings in Experiment 1 were influenced by auditory cues in the voice or visual cues in the body and the face.

Participants

We again recruited 57 participants in Experiment 2 so that we had an identical participant number as in Experiment 1 (participants in Experiment 2 had not been involved in Experiment 1). Data collection was terminated after reaching 57 complete datasets ($N = 57$, 38 male and 19 female; $M_{\text{years}} = 30.23$; $SD = 10.11$). All participants gave written informed consent.

Procedure

Everything was identical to Experiment 1 with the only exception that videos were presented without any sound.

Results

The descriptive statistics are summarized in Table 2. The data source file can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

A 2 (Early vs. Late) $\times$ 2 (Depleted vs. Non depleted) ANOVA on the mean ($M_{\text{early-depleted}} = 5.26$, $SD = 0.87$; $M_{\text{early-non-depleted}} = 5.41$, $SD = 0.79$; $M_{\text{late-depleted}} = 4.64$, $SD = 0.93$; $M_{\text{late-non-depleted}} = 4.84$, $SD = 0.94$) of the seven dependent variables ($\alpha = .98$) revealed a significant main effect of measurement time on overall perceived self-control capacity of the targets, $F(1, 56) = 140.103$, $p < .001$, $\eta_p^2 = .714$, demonstrating that the participants rated the targets lower on perceived self-control toward the end of the Stroop task as compared to the beginning. A significant main effect of depletion group, $F(1, 56) = 21.926$, $p < .001$, $\eta_p^2 = .281$, indicated that participants rated targets as possessing less perceived self-control capacity in the depletion group as compared to the non-depletion group. The interaction between measurement time and depletion group was not significant, $F(1, 56) = 0.519$, $p = .474$, $\eta_p^2 = .009$. Further analyses on the individual perceived self-control items can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

Table 2. Descriptive statistics (*M* and *SD* in brackets) as a function of measurement time and depletion group on the individual dependent variables in Experiment 2 (without sound). Low values always reflect low levels of self-control on the individual items

	Early		Late	
	Depleted	Non-depleted	Depleted	Non-depleted
Awake (1 = <i>not at all</i> ; 8 = <i>very</i>)	5.57 (0.91)	5.72 (0.84)	4.90 (0.97)	5.19 (0.99)
Balanced (1 = <i>not at all</i> ; 8 = <i>very</i>)	5.26 (0.81)	5.34 (0.74)	4.80 (0.92)	4.88 (0.93)
Lack of drive (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.18 (0.99)	5.46 (0.88)	4.48 (1.00)	4.83 (1.00)
Exhausted (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.19 (0.90)	5.31 (0.89)	4.56 (0.98)	4.67 (0.99)
Mentally fatigued (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.20 (0.93)	5.32 (0.89)	4.56 (1.00)	4.67 (1.10)
Depleted (1 = <i>very</i> ; 8 = <i>not at all</i>)	5.18 (0.96)	5.32 (0.91)	4.56 (1.00)	4.96 (1.00)
Will power (1 = <i>not at all</i> ; 8 = <i>very</i>)	5.23 (0.96)	5.43 (0.85)	4.63 (1.00)	4.90 (0.97)

Discussion

The primary aim of Experiment 2 was not only to replicate the findings of Experiment 1 in another sample, but also to identify the role of visual apart from auditory information in recognizing others' self-control capacity. Experiment 2 supported our previous findings insofar that depleted participants were perceived as having less self-control capacity during the Stroop task than non-depleted participants. Moreover, the participants rated the target persons again as having less self-control capacity at the end compared to the beginning of the self-control demanding Stroop task.

While the main effects replicated the findings of Experiment 1, the expected interaction between depletion group and time during the Stroop task was not significant. One plausible reason why the interaction was not significant in Experiment 2 compared to Experiment 1 might be that auditory cues in the voice bring additional information that allow even more fine-grained perception of others' self-control states. Nevertheless, the participants of Experiment 2 were still able to correctly identify targets' general state of self-control capacity as well as changes in self-control capacity over time, by exclusively observing the targets' nonverbal behavior (i.e., visual information about facial and bodily expression).

In a third experiment, we aimed at replicating the finding that ego-depletion is detectable in nonverbal behavior, again. This time, however, we used a different measure for the dependent variable as ego-depletion has been linked to other internal states (including fatigue and negative affect; e.g., Inzlicht & Schmeichel, 2012; Inzlicht et al., 2014; Marcora, Staiano, & Manning, 2009) in order to gain a better understanding of the nature of the expressions in the stimulus material used in this study.

Experiment 3

Experiment 3 was similar to Experiment 1 only that the dependent variables were altered as ego-depletion has been

linked to other internal states including fatigue and negative affect (Hagger et al., 2010) which both have been shown to be detectable from nonverbal expressions (Carney et al., 2007; Sundelin et al., 2013). By changing the dependent variables we aimed to avoid mono-method bias by applying a different rating instrument of momentary self-control capacity. Further, we attempted to gain understanding if the accurate ratings of Experiments 1 and 2 are indicative of a distinct expression of self-control capacity or can be considered a blend of expressions that are associated with internal states that have been linked to self-control capacity.

Experiment 3 used fatigue rating scales from an established instrument (i.e., Watson & Clark, 1994, PANAS-X) as an indicator of momentary self-control capacity. In a recent meta-analysis, Hagger et al. (2010) found a significant association between fatigue and ego-depletion across different studies, leading them to suggest that fatigue may serve as an indicator of the increased demands that self-control places on individuals. More specifically, theoretical and empirical findings suggest that people experience subjective fatigue when exerting mental effort (Cameron, 1973). In this respect, depleted self-control capacity has been shown to coincide with subjectively felt fatigue (Finkel et al., 2006; Friese, Hofmann, & Wanke, 2008; Stewart, Wright, Hui, & Simmons, 2009) and physiological indicators of generalized fatigue (Segerstrom & Nes, 2007; Wright et al., 2007, Wright, Stewart, & Barnett, 2008). These findings support the idea that self-control tasks are more demanding relative to the control condition in the sequential-task paradigm and are therefore more taxing on individuals' resources and therefore can lead to fatigue. Hence, we reasoned that fatigue might serve as an observable indicator of the increased demands that self-control tasks place on individuals. As previous research has suggested that similar to the nonverbal communication of emotions (Darwin, 1872; Ekman, 1992), the internal state of fatigue or sleepiness communicates a motivational drive to rest (Fridlund, 1994; Sundelin et al., 2013), we attempted to test if a brief period of self-control effort would result in higher observer ratings of generalized fatigue.

Second, negative affect has also been discussed in relation to ego-depletion, albeit with less consistent associations across studies (Hagger et al., 2010). It has been suggested that tasks requiring self-control are demanding and frustrating and therefore may induce a negative affective state (Inzlicht et al., 2014; Leith & Baumeister, 1996; Tice, Bratslavsky, & Baumeister, 2001). However, studies including measures of negative affect as an additional dependent variable in the sequential-task paradigm have provided mixed results. While some studies did not find a relationship between ego-depletion and negative affect (e.g., Baumeister et al., 1998; Bruyneel, Dewitte, Franses, & Dekimpe, 2009; Muraven et al., 1998), other studies have reported significant post depletion increases in negative affect (after controlling for baseline) relative to non-depleted controls (Ciarocco, Sommer, & Baumeister 2001; Stewart et al., 2009). Importantly, the emotion-shift hypothesis (Inzlicht et al., 2014) predicts that negative affect should increase over time when exerting self-control effort. Hence, we additionally tested if targets would be rated differently in terms of negative affect (for the distinction between fatigue and negative affect, see Hockey, 2013).

Of importance to Experiment 3, we choose three items from the PANAS-X (Watson & Clark, 1994; German translation by Grühn, Kotter-Grühn, & Röske, 2010) assessing fatigue that were arguably distinct from items used in Experiments 1 and 2: drowsy, tired, and sluggish. We used these three items as an indicator of fatigue (Watson & Clark, 1994). Further, we chose five items from the PANAS-X, assessing negative affect: distressed (bedrückt), ashamed (beschämt), irritable (reizbar), upset (verärgert), afraid (ängstlich). We used these five items as an indicator of a person's negative affective state.

Participants

We again recruited 57 participants in Experiment 3 so that we had an identical participant number compared to our previous experiments (the participants of Experiment 3 did not participate in the two previous experiments). Data collection was terminated after reaching 57 complete data-sets ($N = 57$, 23 male and 34 female; $M_{\text{years}} = 30.49$; $SD = 9.42$). All participants gave written informed consent.

Procedure

Everything was identical to Experiment 1 with the only exception that the dependent variables were altered. We used three items from the PANAS-X (Watson & Clark, 1994; German translation by Grühn et al., 2010) assessing fatigue: drowsy (dösig), tired (müde), sluggish (träge); and

five items assessing negative affect: distressed (bedrückt), ashamed (beschämt), irritable (reizbar), upset (verärgert), and afraid (ängstlich).

Results

The descriptive statistics are summarized in Table 3. The data source file can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d

A 2 (Early vs. Late) $\times$ 2 (Depleted vs. Non depleted) ANOVA on the mean of the three fatigue items ($M_{\text{early-depleted}} = 3.24$, $SD = 0.97$; $M_{\text{early-non-depleted}} = 3.05$, $SD = 0.98$; $M_{\text{late-depleted}} = 3.90$, $SD = 1.04$; $M_{\text{late-non-depleted}} = 3.62$, $SD = 1.05$) revealed a significant main effect of measurement time on overall perceived fatigue ($\alpha = .94$) of the targets, $F(1, 56) = 121.924$, $p < .001$, $\eta_p^2 = .685$. Participants rated the targets substantially higher on perceived fatigue toward the end of the Stroop task as compared to the beginning. A significant main effect of depletion group, $F(1, 56) = 15.885$, $p < .001$, $\eta_p^2 = .221$, indicated that participants rated targets as being more fatigued in the depletion group as compared to the non-depletion group. The interaction between measurement time and depletion group was not significant, $F(1, 56) = 0.829$, $p = .366$, $\eta_p^2 = .015$. Further analyses on the individual items can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

A 2 (Early vs. Late) $\times$ 2 (Depleted vs. Non depleted) ANOVA on the mean of the five negative affect items ($M_{\text{early-depleted}} = 2.19$, $SD = 0.85$; $M_{\text{early-non-depleted}} = 2.27$, $SD = 0.79$; $M_{\text{late-depleted}} = 2.36$, $SD = 0.88$; $M_{\text{late-non-depleted}} = 2.40$, $SD = 0.89$) revealed a significant main effect of measurement time on overall perceived negative affect ($\alpha = .95$) of the targets, $F(1, 56) = 21.327$, $p < .001$, $\eta_p^2 = .276$. Participants rated the targets substantially higher on perceived negative affect toward the end of the Stroop task as compared to the beginning. The main effect of depletion group, $F(1, 56) = 3.046$, $p = .086$, $\eta_p^2 = .052$, and the interaction between measurement time and depletion group, $F(1, 56) = 0.497$, $p = .484$, $\eta_p^2 = .009$, on negative affect were not significant. Further analyses on the individual negative affect items can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

Discussion

The aim of Study 3 was to better understand if the findings of Experiments 1 and 2 are indicative of a distinct expression of self-control capacity or can be considered a blend of expressions that are associated with internal states (fatigue and negative affect) that have been linked to

Table 3. Descriptive statistics (*M* and *SD* in brackets) as a function of measurement time and depletion group on the individual dependent variables in Experiment 3

	Early		Late	
	Depleted	Non-depleted	Depleted	Non-depleted
<i>Fatigue</i>				
Drowsy (1 = <i>not at all</i> ; 8 = <i>very</i>)	3.22 (1.00)	2.95 (1.00)	3.82 (1.10)	3.54 (1.10)
Tired (1 = <i>not at all</i> ; 8 = <i>very</i>)	3.29 (1.10)	3.17 (1.10)	4.04 (1.10)	3.78 (1.10)
Sluggish (1 = <i>not at all</i> ; 8 = <i>very</i>)	3.21 (1.00)	3.01 (1.10)	3.83 (1.10)	3.54 (1.10)
<i>Negative affect</i>				
Distressed (1 = <i>not at all</i> ; 8 = <i>very</i>)	2.62 (1.00)	2.42 (1.00)	2.85 (1.1)	2.79 (1.10)
Ashamed (1 = <i>not at all</i> ; 8 = <i>very</i>)	2.33 (0.95)	2.54 (0.95)	2.50 (1.1)	2.40 (0.95)
Irritable (1 = <i>not at all</i> ; 8 = <i>very</i>)	2.14 (0.97)	2.30 (0.83)	2.32 (0.99)	2.47 (0.97)
Upset (1 = <i>not at all</i> ; 8 = <i>very</i>)	1.90 (0.86)	2.04 (0.77)	2.00 (0.85)	2.21 (0.93)
Afraid (1 = <i>not at all</i> ; 8 = <i>very</i>)	1.98 (0.91)	2.04 (0.86)	2.15 (0.93)	2.11 (1.00)

self-control capacity (Hagger et al., 2010). The results revealed that depleted compared to non-depleted target people were indeed rated as being more fatigued based on their nonverbal behavior. Thus, we could replicate the results of the previous two experiments with different items assessing fatigue. This suggests that the state of ego-depletion does not have a distinct expression (e.g., as has been argued for some basic emotions; Ekman & Cordaro, 2011). Rather, it is more likely that nonverbal expressions that coincide with ego-depletion can be considered a complex blend of expressions associated with the internal states of fatigue and some forms of negative affect (although the main effect for depletion group on negative affect was not significant). In this respect, the findings are supportive of the Behavioral Ecology Theory of facial expressions (Fridlund, 1994).

As in Experiment 2, the interaction effect was not statistically significant for fatigue and negative affect. Thus, while we found that depleted targets appeared to be more fatigued to perceivers in general, this effect was not significantly stronger at the end compared to the beginning of the Stroop task.

While perceptions of general fatigue were consistently associated with the state of ego-depletion, the pattern for negative affect was less consistent (see Table 3) and effect sizes were considerably smaller for the factor measurement time. Although, there were some relative differences between the experimental groups, target people were rated low on negative affect on the 8-point negative affect items (see Table 3), also in the depletion group at the end of the Stroop task. Hagger et al. (2010) expected ego-depletion to be related to negative affect given the highly aversive and effortful nature of self-control tasks (and they actually found this relationship in their meta-analysis). Our results indicate, however, that the term negative affect may be

too unspecific in this regard—at least in terms of the visible part of affect which is supportive of recent theorizing on affect (Cowen & Keltner, 2017). In sum, negative affect was not as clearly detectable in the nonverbal behavior of depleted target people as this was the case for items that have been linked to self-control capacity (Experiment 1 and 2) and fatigue (Experiment 3).

Taken together the results of Experiment 3 suggest that nonverbal expressions that coincide with ego-depletion most likely do not have a distinct expression, but rather are associated with various nonverbal cues that are associated with momentary levels of self-control capacity, fatigue, and some negative affective states. However, we acknowledge that this interpretation of the results of Experiment 3 warrants some caution as previous research has indicated that the specific rating labels in nonverbal behavior research influence observers' judgments. For example, nonverbal behavior previously thought of as representing pride was better characterized as a distinct expression of triumph, when participants were given different response options (Matsumoto & Hwang, 2012). In this respect, it would have been interesting to combine the self-control capacity measure with the fatigue and negative affect items in one study. However, pilot testing that implemented this approach revealed that experimental sessions lasted too long (over 2 hr) and participants lost interest and motivation.

In summary, it is not clear what specific behavioral changes are associated with the internal state of ego-depletion and informed observer ratings in Experiments 1–3. In a step toward identifying objectively observable cues that targets showed and perceivers used to inform their ratings in Experiments 1–3, we conducted a final study, in which we analyzed the stimulus material with automated facial coding software.

Experiment 4

According to a recent review of nonverbal communication research, the face can be considered the single most prominent nonverbal channel as it is the most intricate (Matsumoto & Hwang, 2013, p. 15): “It is the most complex signaling system in our body. [...] And arguably it is the seat of the greatest amount of information that is conveyed nonverbally. That’s why we have ‘face-to-face’ interactions.” Similarly, Cozolino (2006, p. 154) states “the faces of others may be the single most important source of information in our world”. These statements highlight the critical role of the human face in coordinating social interactions. Internal states have been linked to distinct cues, including movements of the facial muscles (Ekman, 2016a for an overview). In this respect, certain emotions (e.g., Ekman & Cordaro, 2011 for a review), but also the state of fatigue (Sundelin et al., 2013) have been shown to be associated with distinct facial features and movements. To examine whether observers’ impressions as captured in Experiments 1–3 can be brought down to (changes in) objectively identifiable facial expressions, we analyzed the stimulus material of Experiments 1–3 with the automated facial coding software FaceReader (Noldus Information Technology, Wageningen, The Netherlands).

Using Active Appearance Modelling (Van Kuilenburg, Wiering, & den Uyl, 2005), FaceReader detects the activation intensity of 20 distinct muscle groups in the face (i.e., “action units”), which are used to generate facial expression (Ekman & Friesen, 1978). Subsequently, whole-face expressions are classified in seven basic emotions, following Ekman and Friesen’s Facial Action Coding System (for validity of this classification procedure, see Lewinski, den Uyl, & Butler, 2014). In addition, a combination of action unit intensity and intensity of positive versus negative emotion is used to infer general levels of valence and arousal (i.e., based on Russell’s circumplex model of affect; Russell, 1980). Because Experiments 1–3 indicated that nonverbal expressions of ego-depletion are likely to reflect a mixture of nonverbal cues that are associated with low momentary levels of self-control (e.g., fatigue, negative affect), our analysis of participants’ facial expression was largely explorative. As such, we analyzed the stimulus material at the level of individual action units, expression of basic emotions, as well as overall valence and arousal. Furthermore, because observers may base their impressions of a target participant on the average expression that is displayed within each 10-second video clip, as well as on changes in expression over time, for each dependent variable, we performed both static face analyses and moving face analyses. No a priori hypotheses were formulated.

Procedure

Each video clip was processed frame-by-frame (at a frequency of 29.97 Hz) with FaceReader 7.0 software (Noldus Information Technology, Wageningen, The Netherlands). A general face model was used as reference, and output was selected to include the intensity of 20 distinct action units, expression of seven basic emotions (i.e., happy, sad, angry, surprised, scared, disgusted, and neutral), and general levels of valence and arousal. In addition, head orientation was measured in three dimensions. Figure 1 provides an impression of main features of the FaceReader analysis. For more information about FaceReader’s methodology, see <http://info.noldus.com/free-white-paper-on-facereader-methodology/>.

Following the initial FaceReader analysis, four participants had to be excluded because FaceReader was unable to reliably detect a face from the videos in one or both experimental conditions (i.e., early and/or late). Full data were available for 33 participants (i.e., 15 depleted and 18 non-depleted). For the static face analyses, average values for each dependent variable were calculated for each participant in each condition. For the moving face analyses, the original values were transformed to express the rate-of-change in a specific variable over time using the formula: $v = \Delta x / t$ (where “ Δx ” equals the difference in a specific variable’s value from one video frame to the next, and “ t ” equals the time covered by each video frame [i.e., 29.97 Hz; 33 ms]), and subsequently transforming the values into absolute values (i.e., not containing “–” or “+”) before calculating averages for each participant in each condition. This way the analyses considered the average magnitude of change over time (i.e., fast changes vs. slow changes, measured in intensity-units per second) and not the direction of the change (i.e., increase or decrease).

Results

Descriptive statistics for the static face analyses are summarized in Table 4. As can be seen in Table 4, action unit intensity was generally low and emotional expression was largely neutral, reflecting low levels of arousal and neutral valence, respectively. All data source files can be found at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d

A 2×2 MANOVA for action unit intensity indicated no significant main effect of time (early vs. late; $V = .616$, $F(17, 15) = 1.418$, $p = .251$, $\eta_p^2 = .616$), no significant main effect of group (depletion vs. non-depletion; $V = .565$, $F(17, 15) = 1.146$, $p = .399$, $\eta_p^2 = .565$), and no significant interaction ($V = .451$, $F(17, 15) = 0.724$, $p = .741$, $\eta_p^2 = .451$). Similarly, 2×2 MANOVAs for emotional expression,

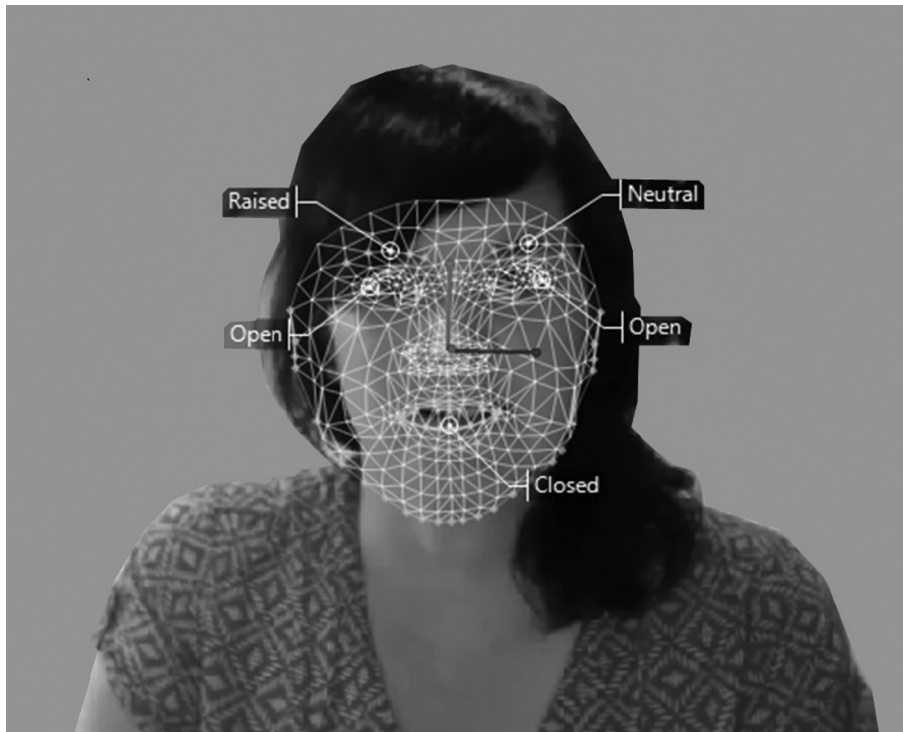


Figure 1. Screenshot of the FaceReader analysis of one of the participants, indicating overlay of face model, action unit activation (e.g., brow raiser, eyes open, mouth closed), and head orientation in three dimensions (blue line [X-axis], red line [Y-axis], green line [Z-axis]).

valence and arousal, and head orientation, did not show significant main effects or interactions (all $p > .136$).

Descriptive statistics for the moving face analyses are summarized in Table 5.

A 2×2 MANOVA for action unit intensity indicated no significant main effect of time, $V = .637$, $F(17, 15) = 1.549$, $p = .200$, $\eta_p^2 = .637$; no significant main effect of group, $V = .447$, $F(17, 15) = 0.712$, $p = .751$, $\eta_p^2 = .447$; and no significant interaction, $V = .529$, $F(17, 15) = 0.989$, $p = .513$, $\eta_p^2 = .529$. Similarly, 2×2 MANOVAs for emotional expression and head orientation also did not show significant main effects or interactions (all p 's $> .211$). The 2×2 MANOVA for valence and arousal, however, did show a significant multivariate main effect of time, $V = .206$, $F(2, 30) = 3.889$, $p = .031$, $\eta_p^2 = .206$. Follow-up univariate ANOVAs indicated that the rate of change in both valence and arousal became slower over time (i.e., was significantly smaller in the late condition than in the early condition), with $F(1, 31) = 6.052$, $p = 0.020$, $\eta_p^2 = .163$ and $F(1, 31) = 6.149$, $p = 0.019$, $\eta_p^2 = .166$. No multivariate main effects of group or interaction were observed (all $p > .539$).

Discussion

The aim of Experiment 4 was to test whether the expression of ego-depletion can be reduced to (changes in) objectively identifiable features of the face that observers' might use to form impressions. The FaceReader analyses did not

reveal significant differences in average expression intensities across experimental conditions, but did show significant differences in facial dynamics for the factor time, that is, slower changes in expression intensity. Stated differently, at the beginning and end of the self-control demanding Stroop task, participants showed similar facial expressions but, at the end of the task, changes in facial expressions (i.e., valence, arousal) occurred at a significantly slower pace. Therefore, a significant effect was only observed in the "moving" face analyses for the factor time and not in the "static" face analyses, suggesting that time may be an important factor in perceiving subjective states from facial expression (Van der Schalk, Hawk, Fischer, & Doosje, 2011). The fact that significant effects were not observed in the static face analysis (i.e., reflecting average levels of expression intensity) is not in line with a recent study that revealed distinct facial features reflecting fatigue (e.g., Sundelin et al., 2013). Note, however, that in this study, the manipulations of fatigue concerned a complete night of sleep deprivation, which is arguably different and more intense than the emotional suppression and time-on-task manipulations that were employed in the current study (see Experiment 1). Of further importance, significant effects were not evident in distinct action units or emotional expressions, but only in our "aggregate" measures of valence and arousal. This observation supports the notion from Experiment 3 that nonverbal expressions of ego-depletion may not be restricted to distinct facial features, but may rather be associated with a combination of cues

Table 4. Descriptive statistics (*M* and *SD* in brackets) as a function of measurement time (early vs. late) and depletion group (depletion vs. non-depletion) on the individual static face variables in Experiment 4

	Early		Late	
	Depleted	Non-depleted	Depleted	Non-depleted
<i>Action unit intensity</i>				
Inner brow raiser	0.00 (0.00)	0.02 (0.07)	0.01 (0.05)	0.00 (0.00)
Outer brow raiser	0.00 (0.01)	0.05 (0.15)	0.01 (0.02)	0.03 (0.11)
Brow lowerer	0.02 (0.05)	0.04 (0.13)	0.03 (0.10)	0.05 (0.19)
Upper lid raiser	0.04 (0.12)	0.01 (0.03)	0.00 (0.01)	0.00 (0.01)
Cheek raiser	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.02)
Lid tightener	0.02 (0.09)	0.00 (0.02)	0.02 (0.09)	0.03 (0.08)
Nose wrinkler	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Upper lip raiser	0.00 (0.00)	0.01 (0.03)	0.00 (0.02)	0.01 (0.03)
Lip corner puller	0.02 (0.08)	0.05 (0.10)	0.00 (0.01)	0.03 (0.10)
Dimpler	0.01 (0.02)	0.00 (0.01)	0.00 (0.01)	0.00 (0.00)
Lip corner depressor	0.01 (0.04)	0.00 (0.00)	0.01 (0.04)	0.00 (0.00)
Chin raiser	0.01 (0.02)	0.02 (0.08)	0.00 (0.01)	0.03 (0.10)
Lip pucker	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Lip stretcher	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Lip tightener	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.01)
Lip pressor	0.01 (0.02)	0.02 (0.05)	0.00 (0.00)	0.02 (0.05)
Lips part	0.05 (0.07)	0.03 (0.07)	0.03 (0.08)	0.03 (0.09)
Jaw drop	0.06 (0.15)	0.01 (0.04)	0.04 (0.11)	0.02 (0.08)
Mouth stretch	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Eyes closed	0.08 (0.18)	0.01 (0.04)	0.07 (0.20)	0.02 (0.06)
<i>Emotional expression</i>				
Neutral	0.69 (0.20)	0.69 (0.16)	0.73 (0.17)	0.73 (0.18)
Happy	0.12 (0.15)	0.17 (0.16)	0.07 (0.08)	0.14 (0.16)
Sad	0.09 (0.11)	0.09 (0.11)	0.13 (0.18)	0.08 (0.12)
Angry	0.07 (0.05)	0.07 (0.05)	0.06 (0.04)	0.07 (0.05)
Surprised	0.08 (0.10)	0.05 (0.07)	0.07 (0.08)	0.06 (0.07)
Scared	0.03 (0.03)	0.02 (0.02)	0.03 (0.03)	0.02 (0.01)
Disgusted	0.02 (0.02)	0.03 (0.07)	0.01 (0.01)	0.02 (0.02)
<i>Valence</i>	−0.00 (0.19)	0.05 (0.17)	−0.08 (0.19)	0.02 (0.18)
<i>Arousal</i>	0.36 (0.11)	0.38 (0.14)	0.33 (0.08)	0.35 (0.16)
<i>Head orientation</i>				
Y-axis	8.75 (9.73)	4.44 (12.73)	4.08 (14.38)	5.33 (9.65)
X-axis	2.22 (5.11)	2.13 (2.22)	1.11 (5.06)	2.59 (3.40)
Z-axis	0.74 (4.49)	0.95 (4.65)	2.75 (5.03)	−1.03 (5.17)

Note. Values for action unit intensity range from 0 (= no activity) to 1 (= maximal activity), values for emotional expression range from 0 (= absent) to 1 (= fully present), values for valence range from −1 (= negative) to +1 (= positive), and values for arousal range from 0 (= not active) to 1 (= active). Head orientation is expressed in degrees (°).

that together reflect low levels of activation and a lack of emotionality (i.e., neutral emotional expression). The current analysis adds to this impression, by suggesting that it is not the lack of expression intensity that characterizes individuals with low levels of self-control, but rather a slowing down of changes in expression intensity over time.

Another important observation is that in our analysis of objectively identifiable facial features, we only observed a significant main effect of time (i.e., early vs. late) and—unlike the subjective perceptions were captured in

Experiments 1–3 – did not observe a significant main effect of depletion group (depleted vs. non-depleted), or interaction between depletion group and time. In addition, effect sizes were generally a lot smaller. This difference indicates that although a slowing down of changes in facial expression may inform the perception of ego-depletion (at least for the time factor), participants in Experiments 1–3 are likely to have used a variety of other cues (e.g., skin color, body posture) that together “sharpened” their impression. Finally, an important limitation that has to be acknowledged in

Table 5. Descriptive statistics (*M* and *SD* in brackets) as a function of measurement time (early vs. late) and depletion group (depletion vs. non-depletion) on the individual moving face variables in Experiment 4

	Early		Late	
	Depleted	Non-depleted	Depleted	Non-depleted
<i>Action unit intensity</i>				
Inner brow raiser	0.00 (0.00)	0.02 (0.07)	0.03 (0.09)	0.00 (0.01)
Outer brow raiser	0.01 (0.03)	0.03 (0.07)	0.02 (0.06)	0.02 (0.07)
Brow lowerer	0.04 (0.08)	0.02 (0.05)	0.02 (0.05)	0.01 (0.04)
Upper lid raiser	0.03 (0.06)	0.02 (0.08)	0.01 (0.04)	0.01 (0.03)
Cheek raiser	0.00 (0.00)	0.00 (0.01)	0.00 (0.00)	0.01 (0.03)
Lid tightener	0.03 (0.11)	0.01 (0.02)	0.03 (0.11)	0.02 (0.06)
Nose wrinkler	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Upper lip raiser	0.00 (0.01)	0.01 (0.04)	0.01 (0.05)	0.01 (0.05)
Lip corner puller	0.03 (0.09)	0.07 (0.12)	0.01 (0.02)	0.02 (0.05)
Dimpler	0.01 (0.06)	0.00 (0.02)	0.01 (0.02)	0.00 (0.00)
Lip corner depressor	0.02 (0.07)	0.00 (0.00)	0.03 (0.09)	0.00 (0.00)
Chin raiser	0.02 (0.04)	0.04 (0.12)	0.02 (0.04)	0.04 (0.09)
Lip puckerer	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Lip stretcher	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Lip tightener	0.00 (0.00)	0.00 (0.00)	0.01 (0.05)	0.00 (0.00)
Lip pressor	0.02 (0.06)	0.03 (0.09)	0.00 (0.01)	0.03 (0.08)
Lips part	0.07 (0.07)	0.05 (0.08)	0.02 (0.05)	0.02 (0.05)
Jaw drop	0.03 (0.07)	0.02 (0.05)	0.04 (0.08)	0.01 (0.04)
Mouth stretch	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Eyes closed	0.07 (0.13)	0.03 (0.07)	0.05 (0.10)	0.04 (0.10)
<i>Emotional expression</i>				
Neutral	0.09 (0.04)	0.10 (0.05)	0.09 (0.04)	0.08 (0.04)
Happy	0.05 (0.06)	0.10 (0.10)	0.04 (0.06)	0.06 (0.07)
Sad	0.05 (0.05)	0.04 (0.04)	0.06 (0.04)	0.04 (0.05)
Angry	0.03 (0.03)	0.03 (0.02)	0.03 (0.02)	0.03 (0.02)
Surprised	0.05 (0.04)	0.03 (0.03)	0.05 (0.05)	0.03 (0.03)
Scared	0.01 (0.01)	0.01 (0.01)	0.02 (0.02)	0.01 (0.01)
Disgusted	0.01 (0.01)	0.02 (0.05)	0.01 (0.01)	0.01 (0.01)
<i>Valence</i>	0.10 (0.08)	0.13 (0.10)	0.08 (0.05)	0.09 (0.07)
<i>Arousal</i>	0.04 (0.04)	0.04 (0.03)	0.03 (0.02)	0.03 (0.03)
<i>Head orientation</i>				
Y-axis	9.77 (3.98)	12.67 (6.53)	10.31 (6.31)	11.37 (5.63)
X-axis	8.37 (4.13)	8.12 (3.29)	8.55 (3.53)	7.44 (3.18)
Z-axis	3.28 (1.76)	3.80 (1.50)	3.95 (2.77)	3.37 (1.15)

Experiment 4 is the low power of the face reader analyses, which only included 10-second video clips of 15 depleted and 18 non-depleted participants at two measurement points. Future work is necessary to uncover variability in these cues and to directly link this to subjective perceptions.

General Discussion

The present results are the first to show that the internal state of ego-depletion can reliably be detected by untrained

observers based on thin slices of behavior. Not only was the depletion group rated lower on items that have been associated with self-control capacity, but perceived self-control capacity also declined in both groups from the beginning of the Stroop test to the end. Observers perceived targets to have relatively less self-control capacity the more the targets had to strain their self-control during the sequential-task protocol. Findings from Experiment 3 suggest that nonverbal expressions that coincide with ego-depletion most likely do not have a distinct expression, but rather are comprised of various nonverbal cues that are associated with momentary levels of self-control capacity, fatigue, and

some negative affective states. The results of the face reader analyses in Experiment 4 suggest that facial expressions coinciding with ego-depletion are subtle and do not seem to be reflected in average expression intensities of individual facial muscles, but seem to be extractable from facial dynamics. In this respect, slower changes in expressions may have contributed to participants' ratings in Experiments 1–3.

The present pattern of findings lends further support to the notion that humans have evolved to be very well-equipped to accurately detect (Bogdan, 1997) fine-grained nuances from thin-slices of behavior (Ambady et al., 2000). In his book "The Descent of Man", Charles Darwin (1871, p. 492) wrote: "The sight of another person enduring hunger, cold, fatigue revives in us some recollection of these states, which are painful even in the idea." This quote highlights Darwin's idea that humans have evolved to be highly sensitive toward signals informing observers about the internal struggles going on within other group members. In line with this theorizing, observers of the present studies were able to detect if target people had been exerting relatively more self-control when suppressing their emotional reactions while watching an emotional video for ten minutes and subsequently performing the Color-Stroop task. This finding is also in line with Darwin's (1872) anticreationist argument (i.e., that certain muscles were not "designed" by a divine designer for expression) that expressions do not necessarily have to be functional for the expresser as expressing a lack of self-control presumably can be detrimental to the expresser. In this regard, the present findings might be regarded as supportive of the Behavioral Ecology Theory of facial expressions (Crivelli & Fridlund, 2018; Fridlund, 1994) that regards nonverbal expressions as lead signs that guide further social interaction, instead of automatic displays that are contingent to basic emotional affect programs (e.g., Ekman, 2016b). However, future research is needed to investigate how nonverbal displays associated with low or high levels of self-control capacity affect social interactions.

How might the present findings contribute to the ego-depletion debate? First, it is important to note that the present research was not tailored to test competing models of self-control against each other, but instead to address the novel research question of whether the internal state of ego-depletion can be detected from observations of nonverbal behavior. Nevertheless, besides offering novel methodical tools (e.g., manipulation check measures) for advancing the ego-depletion debate (cf. Lurquin & Miyake, 2017), the findings might arguably be more supportive of the motivation/emotion-shift model of self-control (Inzlicht et al., 2014): that is, this model predicts that depleted individuals experience changes in motivation and affect in the sequential-task paradigm that have been shown to be

detectable from a persons' nonverbal behavior (see Ambady et al., 2000 for a review). However, research on nonverbal behavior has shown that observers can make a broad range of meaningful inferences based on short observations of behavior (Ambady & Rosenthal, 1992) and it therefore remains unclear whether observer ratings were informed by differential levels of motivation and affect or some kind of limited resource.

Further, the present results extend previous findings showing that sleep deprivation can reliably be detected from facial cues by observers (Axelsson et al., 2010; Sundelin et al., 2013). However, the cues that people use seem to be different as the analyses of Experiment 4 indicated: that is, not the lack of expression intensity (see Sundelin et al., 2013) characterized individuals with low levels of self-control, but rather a slowing down of changes in expression intensity over time. Hence, subtle dynamical changes in the face contribute to the nonverbal communication of (even very brief) periods of prior self-control exertion. It has been argued that the internal state of fatigue or sleepiness communicates a motivational drive to rest (Fridlund, 1994; Sundelin et al., 2013). Similarly, exerting self-control capacity might signal a drive for resource "repletion" or a lower motivation to engage in further effortful self-control (see Baumeister, 2014; Inzlicht & Schmeichel, 2012 for recent discussions on this topic). Concerning the latter, a recent study by Lindner, Nagy and Retelsdorf (2018) provided evidence for a close interplay between available self-control capacity and the motivation to invest effort in a 140-min achievement test.

The vast majority of studies investigating the recognition of facial expressions have focused on static displays of intense emotions. Recent research has started to highlight the importance of facial dynamics "for deciphering the fleeting subtle expressions that permeate real life situations" (Ambadar, Schooler, & Cohn, 2005, p. 403). This research shows that subtle facial expressions that were not identifiable in static presentations suddenly became apparent in dynamic displays (Ambadar et al., 2005). As the visual system evolved under dynamic conditions (see Gibson, 1966), people have evolved to be highly attuned to motion signals. This theorizing also seems apparent in the present results as the stimulus material did not show significant differences across experimental conditions in average expression intensities of facial muscles, but indicated differences in facial dynamics across the experimental time conditions. In line with a recent review (Krumhuber, Kappas, & Manstead, 2013) emphasizing the importance of dynamic facial information for detecting and distinguishing between subtle facial expressions, Experiment 4 indicates that slower changes in facial expressions coincide with depleted states of self-control capacity and, thus, may have contributed to the accurate

rating differences across experimental stimulus categories in Experiments 1–3.

The present research approach has some notable strengths and weaknesses. We consider it a strength that three studies consistently showed that depleted self-control is communicated nonverbally. While extensive previous research has focused on the intrapersonal consequences of ego-depletion, resulting in a vivid debate on the nature, the mechanisms, and the magnitude of ego-depletion effects (Carter & McCullough, 2013; Dang, 2018; Hagger & Chatzisarantis, 2014), the present research is the first to highlight that the internal state of ego-depletion is socially broadcasted and thereby might have important interpersonal effects. For example, observers recognizing that a person is depleted could reflect this in their own nonverbal behavior and might treat this person differently (e.g., have low expectations in this person's performance on a task). The depleted person is likely to recognize this in the observer and, in turn, might behave accordingly (Jussim, Eccles, & Madon, 1996). We do not want to overestimate potential interpersonal effects of ego-depletion. Kahneman (2011) nicely illustrates the danger of overestimating the effect of variables of investigation within the focusing illusion: "Nothing in life is as important as you think it is while you are thinking about it" (p. 402). However, future research on ego-depletion might benefit from incorporating the present findings on the nonverbal communication of self-control depletion. For example, nonverbal behavior could be included as an additional dependent variable or as a manipulation check measure in future work using the sequential task paradigm.

The main limitation of our research is that we only sampled stimulus material from one study with 36 participants using one version of the sequential-task paradigm. While we found reliable evidence that the internal state of ego-depletion could reliably be detected by untrained observers in every study we did with this stimulus material, future research should conceptually replicate these first findings with different stimulus material. However, it is important to note that the present study design of rating 72 10-second target videos on 7–8 scales (Experiments 1–3) took every participant approximately 1–1.5 hr. Results from pilot testing and recommendations from the thin slices literature (Carney et al., 2007; Murphy et al., 2014) indicated that longer experimental durations are problematic due to attentional and motivational issues of participants. Thus, future research is well advised to sample stimulus material from a larger number of participants using different self-control tasks and randomly sample a subset from the stimulus pool and present this to observers to gain converging evidence for the hypothesis that people leak nonverbal signals that are indicative of how likely they are to show successful acts of self-control. Given the reliability with which people can

detect internal states in others (Ambady et al., 2000) this approach might help in settling some of the controversies in the ego-depletion literature (Hagger et al., 2016; Inzlicht et al., 2014; Lurquin & Miyake, 2017).

In addition, the FaceReader analysis was rather inconclusive (low-statistical power) and has not convincingly settled the question of which nonverbal signals informed observer ratings in Studies 1–3. While the present research indicates that facial dynamics and cues in a person's voice (comparison between Studies 1 and 2) might be informative regarding ego-depletion, future research should scrutinize this research question, as ego-depletion research would benefit from being able to assess objective signals associated with ego-depletion.

In conclusion, depleted self-control capacity is communicated nonverbally among humans. The present research provided first experimental evidence for this, supporting the anecdotal observation that we can tell when young children have limited self-control capacity while performing Walter Mischel's "Marshmallow Test". In this respect, the present research has the potential to open up a new stream of future research on the interpersonal effects of ego-depletion as it is readily detected by observers.

Research Disclosure Statements

The authors declare that 1) (a) the total number of excluded observations and (b) the reasons for making these exclusions have been reported in ESM 1; 2) that all independent variables or manipulations, whether successful or failed, have been reported in the ESM 1; 3) that all dependent variables or measures that were analyzed for this article's target research question have been reported in ESM 1; 4) that (a) how sample size was determined and (b) your data-collection stopping rule have been reported in ESM 1.

Electronic Supplementary Material

The electronic supplementary material is available with the online version of the article at <https://doi.org/10.1027/1864-9335/a000370>.

ESM 1. Detailed results of initial ego-depletion study for the creation of stimulus material.

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Authorship

Philip Furley developed the study concept in collaboration with Sebastian Kohlhaas, Chris Englert, Arne Nieuwenhuys, and Alex Bertrams. Arne Nieuwenhuys performed the FaceReader analyses. Philip Furley, Sebastian Kohlhaas, and Arne Nieuwenhuys performed the data analyses. All authors contributed to the interpretation of the results. Philip Furley drafted the manuscript, and Sebastian Kohlhaas, Chris Englert, Arne Nieuwenhuys and Alex Bertrams provided critical revisions. All authors approved the final version of the manuscript for submission.

Open Data

We have made all study materials available at https://osf.io/86cre/?view_only=c1cc4381077a4aa3978b633e6501b49d.

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